Goal

To characterize the transcriptomic reprogramming of Arabidopsis thaliana exposed to spaceflight under illuminated conditions, with the objective of identifying genes and pathways altered by the combined effects of microgravity and light.

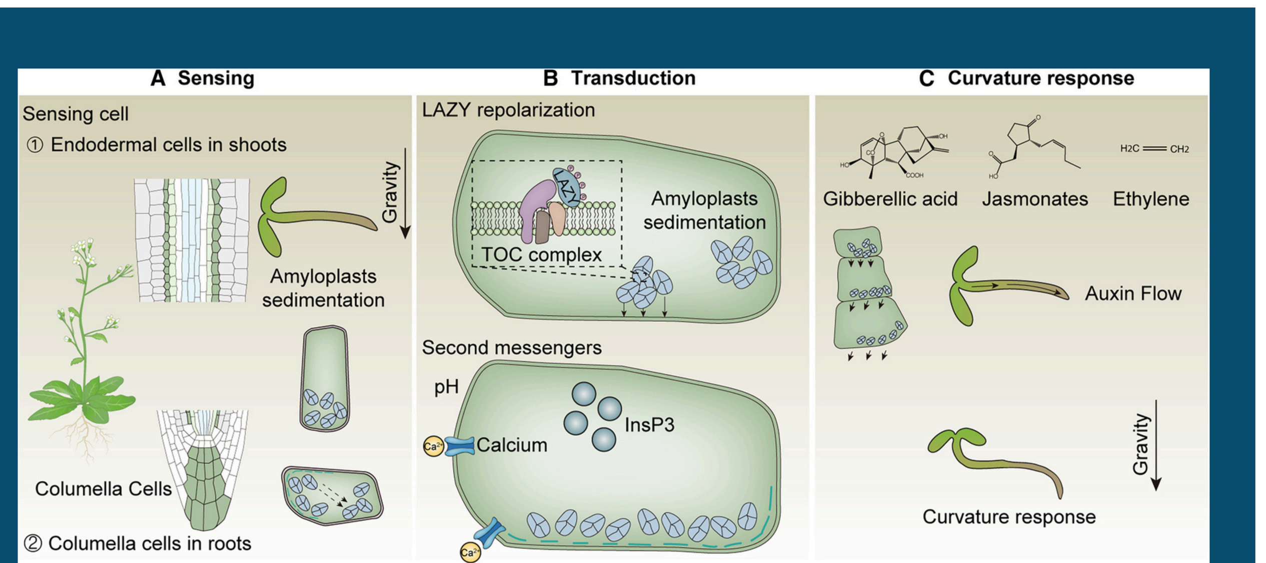
Context

Plants integrate gravitational, hydraulic, and photic cues to regulate stress physiology, hormone signaling, and growth. Spaceflight removes the gravitational vector, altering:

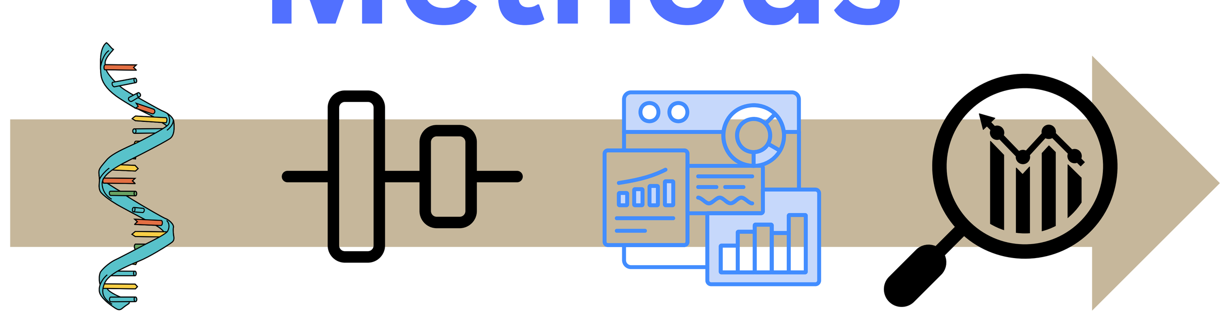
- water redistribution
- intracellular trafficking
- ROS balance
- photosensory signal integration

Previous space biology studies show ABA signaling and osmotic stress responses are consistently affected in microgravity. CARA's root tip analyses showed that phyD substantially reduced the number of differentially expressed genes, indicating distinct organ-level responses to light and microgravity. However, the interaction between light and microgravity, especially in Arabidopsis, remains insufficiently defined.

NASA's OSD-678 Light experiment provides a unique dataset to resolve how lighted microgravity conditions reshape transcriptomic networks.



Methods



Differential expression: DESeq2 comparisons
Space vs. Ground (Light)
Space vs. Ground (Dark)
Gene Set Enrichment: ShinyGO, GO and KEGG, FDR < 0.05 / exploratory FDR < 0.10
Data obtained from NASA GeneLab (GLDS datasets)
Integrative interpretation across light–gravity interaction effects

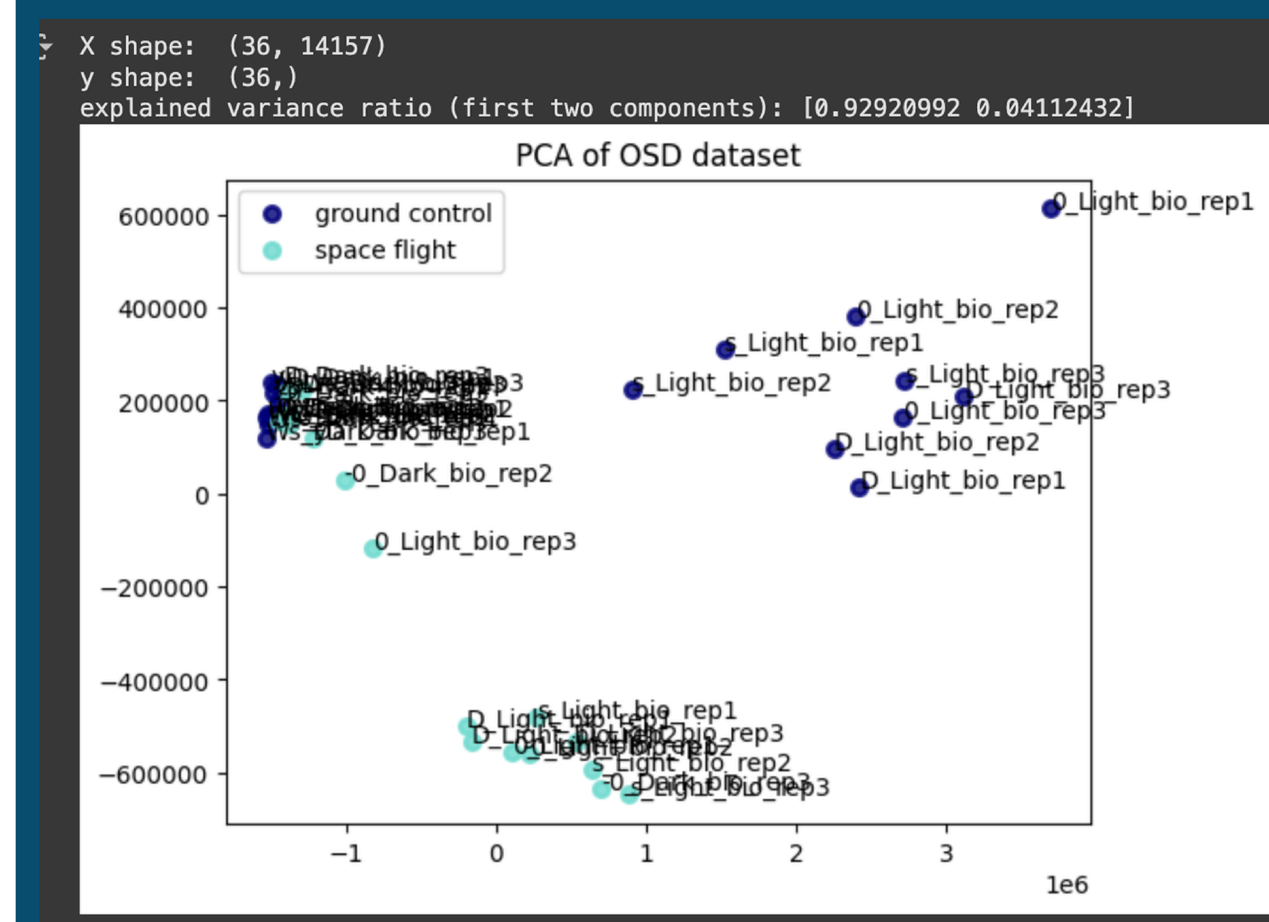
Sample Metadata

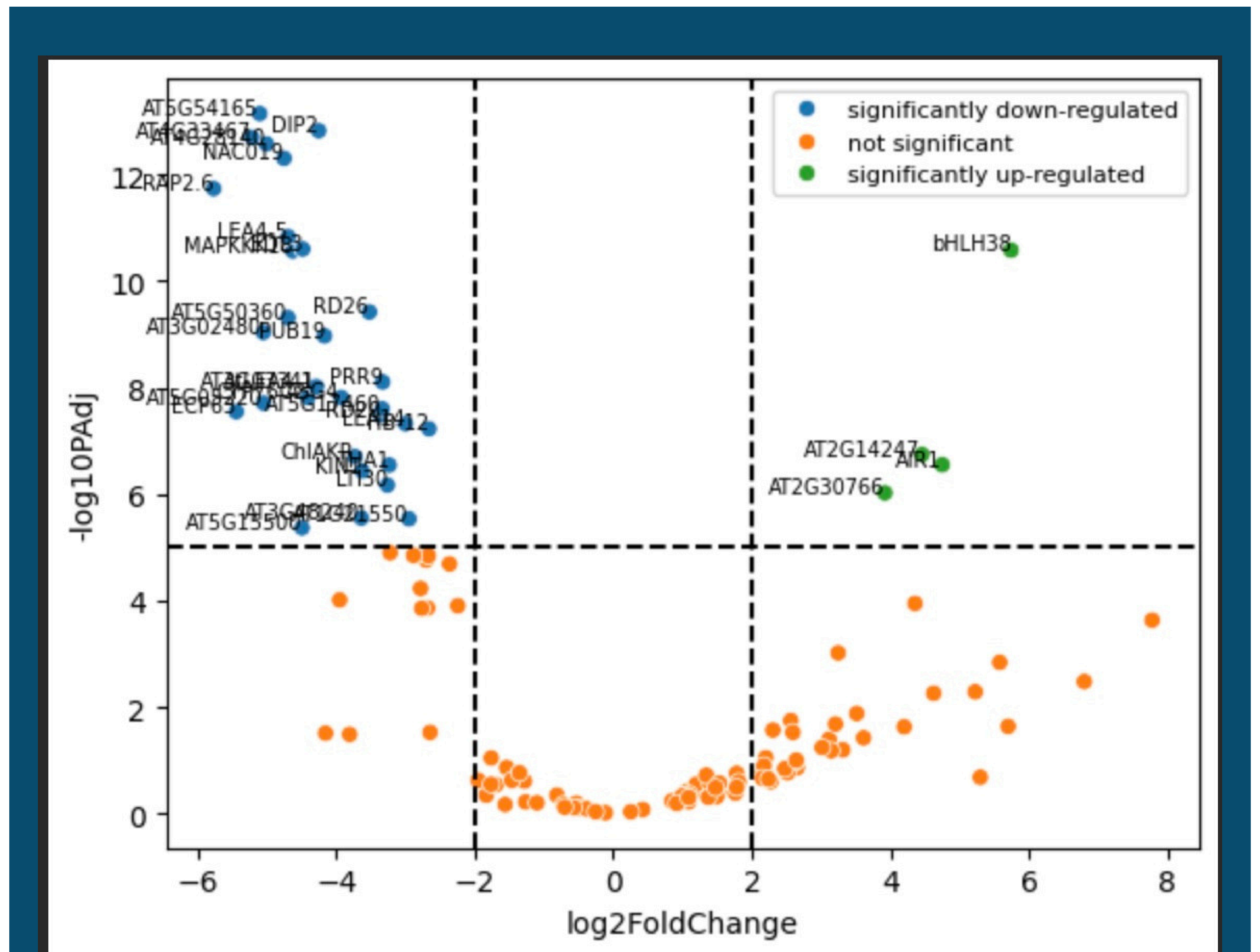
Line	Ecotype	Genotype	Light Condition	Environment
WT-LF	Col-0	WT	Light	Flight
WT-LC	Col-0	WT	Light	Ground
WT-DF	Col-0	WT	Dark	Flight
WT-DC	Col-0	WT	Dark	Ground
phyA Mutants	Col-0	*phyA-deficient*	Light/Dark	Flight/Ground subsets

Findings

1.) Principal Component Analysis

Principal Component Analysis (PCA) revealed strong global transcriptomic structuring across conditions, with PC1 clearly separating spaceflight versus ground samples, indicating that microgravity is the dominant source of variance in the dataset. PC2 separated samples by light versus dark, confirming that photomorphogenic signaling remains a major regulatory factor even in flight. Genotype groups (Col-0, Ws, phyD) clustered tightly within their conditions, demonstrating high replicate quality and minimal batch effects, while phyD samples showed distinct displacement along PC2/PC3, consistent with altered light-response transcriptional programs. Overall, the PCA pattern indicates that spaceflight induces broad, system-level transcriptomic reprogramming rather than isolated gene-level changes.





2.) Differential Expression (DEG)

DESeq2 identified 133 significant DEGs (padj < 0.05, $|\log_2FC| \geq 2$) when comparing spaceflight vs. ground controls under light:

Downregulated genes (strong, coordinated repression of ABA/stress pathways)

Many of the most strongly downregulated transcripts were canonical ABA-dependent, dehydration-responsive, or oxidative-stress–induced genes, including:

- RD20 – a well-established ABA-inducible gene involved in drought and ROS buffering.
- KIN1 – typically induced by cold, ABA, and osmotic stress;
- LTI30 (Low Temperature-Induced 30) – a COR gene tied to cellular dehydration responses;
- LEA4-5 (Late Embryogenesis Abundant) – protects macromolecules under desiccation;

Upregulated genes (nutrient, redox, iron, and RNA-associated regulation)

Upregulated DEGs were enriched for genes involved in metal handling, redox homeostasis, metabolic processing, and nutrient transport, including:

- bHLH38 – a core regulator of iron uptake and homeostasis (FIT partner).
- AT2G14247 – associated with oxidative signaling/stress modulation;
- AIR1 – associated with RNA metabolism and nutrient-responsive pathways;

3.) Gene Set Enrichment

Photosynthesis and Chloroplast-Associated Pathways

- Photosystem I components
- Photosystem II components
- Thylakoid membrane protein complexes
- Light-harvesting complex genes
- Chloroplast redox and electron-transport–associated modules

Phenylpropanoid and Secondary Metabolite Pathways

- Phenylpropanoid biosynthesis
- Lignin precursor biosynthesis
- Flavonoid and anthocyanin-related genes
- Secondary metabolite processing enzymes

Hormone-Associated Gene Sets (Rank-Based Enrichment)

- ABA-associated gene sets (GODA ABA HORMONE UP)
- High-light and salt-treatment–associated sets (HEINRICH HIGH-LIGHT SALT-TREATMENT UP)

Detoxification and Redox Systems

- Glutathione-related pathways
- Peroxidases and oxidoreductases
- Cytochrome P450 superfamily genes
- Glutaredoxin/thioredoxin-related modules

Discussion

Spaceflight with light exposure induced broad transcriptomic reprogramming across all Arabidopsis genotypes. PCA showed that microgravity was the dominant source of variation, with clear separation of flight versus ground samples and consistent genotype-specific substructure, including a distinct shift in phyD that reflects altered photoreceptor signaling. This confirms that the combined microgravity–light environment drives global regulatory changes rather than isolated gene-level disturbances.

Differential expression analysis revealed strong suppression of classical ABA- and dehydration-responsive genes (RD20, KIN1, LTI30, LEA4-5), indicating downregulation of terrestrial osmotic and oxidative stress pathways under microgravity. Conversely, several genes linked to nutrient transport and redox balance (bHLH38, AT2G14247, AIR1) were upregulated, suggesting compensatory activation of metal homeostasis and detoxification processes. These opposing patterns indicate that spaceflight alters how stress perception and resource distribution are transcriptionally prioritized under light.

Pathway-level analyses reinforced these gene-level observations. GSEA identified enrichment of photosystem I/II components, thylakoid membrane complexes, and chloroplast redox modules, reflecting substantial remodeling of photosynthetic and electron-transport systems in microgravity. Additional enrichment in phenylpropanoid biosynthesis, lignin/secondary metabolism, glutathione, oxidoreductases, and P450 pathways indicates enhanced metabolic detoxification and altered carbon allocation. Despite downregulation of many ABA-marker genes, ABA- and high-light–associated gene sets remained enriched, suggesting rewiring rather than simple suppression of stress and hormone networks. Collectively, the data show that plants under light in space adjust stress signaling, nutrient and redox homeostasis, and photosynthetic machinery to maintain functional stability in the absence of gravity-driven environmental cues.

Future is SPACE

S **Systems-level multi-omics integration**

Future studies should combine transcriptomics with proteomics, metabolomics, chromatin accessibility, and hormone profiling to resolve how microgravity reshapes regulatory networks across molecular layers

P **Photoreceptor-specific functional validation**

Given phyD's distinct transcriptomic behavior, CRISPR-engineered photoreceptor mutants and fluorescent reporter lines should be flown to directly test how light-gravity interactions influence signaling and gene expression

A **Adaptive physiology under controlled gravity**

Experiments using centrifuge-equipped chambers on the ISS should examine partial gravity (Moon/Mars levels) to determine thresholds at which ABA suppression, photosynthetic remodeling, and nutrient redistribution occur.

C **Carbon allocation and chloroplast remodeling**

Because GSEA highlighted shifts in phenylpropanoid and photosynthetic pathways, future work should quantify lignin precursors, ROS flux, chloroplast ultrastructure, and secondary metabolite output in microgravity.

E **Environmental simulation**

Ground-based microgravity analogs (clinostats, random positioning machines) and long-duration growth experiments should be expanded to evaluate developmental progression, seed production, and multi-generational stability.

References

- NASA GeneLab Omics Datasets
- NASA Google Colab Analysis Notebooks
- ShinyGO v0.85: GO and KEGG enrichment analysis tool
- All available plant GO gene sets (Arabidopsis thaliana)